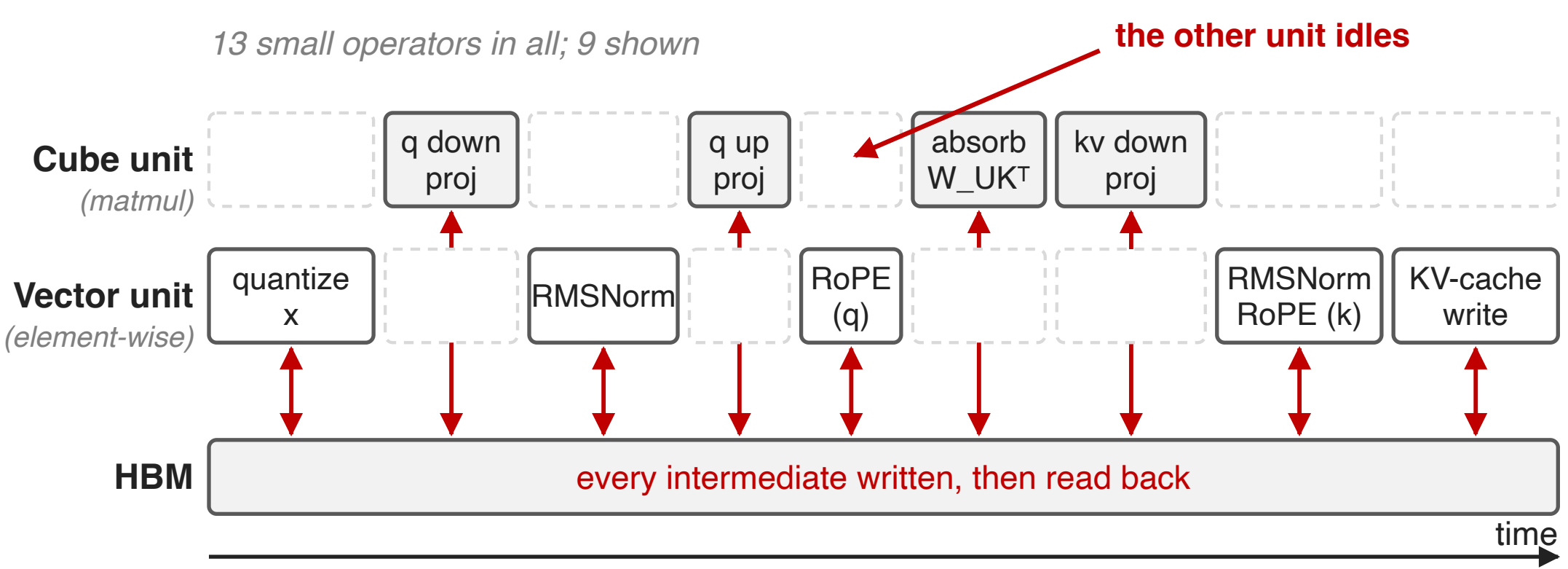

13 small operators in all; 9 shown
the other unit idles
Cube unit
(matmul)
Vector unit
(element-wise)
HBM
q down proj
q up proj
absorb W_UK^T
kv down proj
quantize x
RMSNorm
RoPE (q)
RMSNorm RoPE (k)
KV-cache write
every intermediate written, then read back
time